

This is EC1B1 **Group 40**'s response to the coursework's section 4 and section 5. The group members are: Yiming Li, Mingyun Wang, and Hongyu Ye. The group's allocated country is **Spain**. Graphs/tables/regression results will normally be displayed first for each section, followed by the answers to the coursework's questions at the end of the subsections.

The group's **conclusion** regarding 'Whether exchange rates matter' is that notable exchange rate fluctuations in the history have often come with many confounding events, so it is hard to isolate the impact of exchange rates and judge its importance. In certain cases (e.g., Spain's case following the Bretton Wood's collapse), changes in exchange rate systems might be potentially associated with changes in economic performances, whereas in some cases (e.g., the UK's joining and leaving ERM), exchange rates seemed rather unimportant.

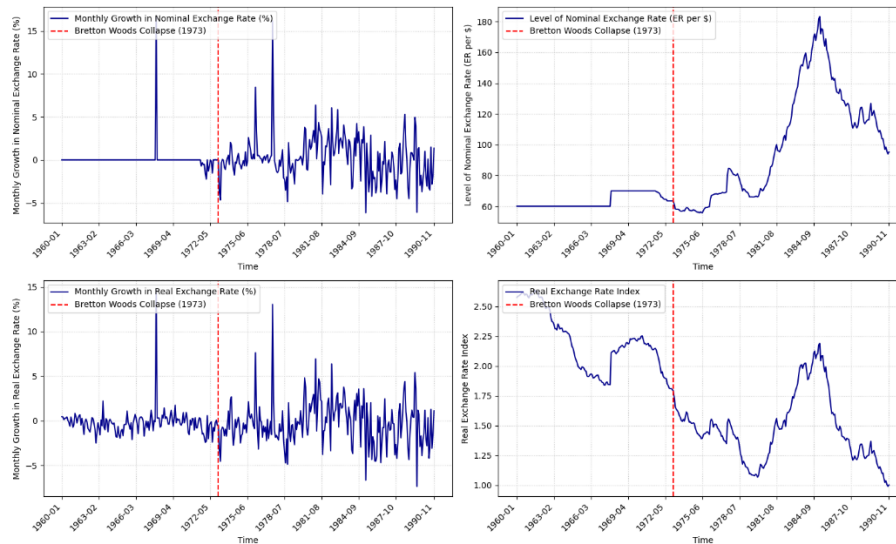
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Section 4: Analysis and Questions

Section 4.1: Exchange Rate and International Reserve Graphs

4.1.1, Exchange Rate Graphs

Spain: Exchange Rate and Index Variations (1960-1990)



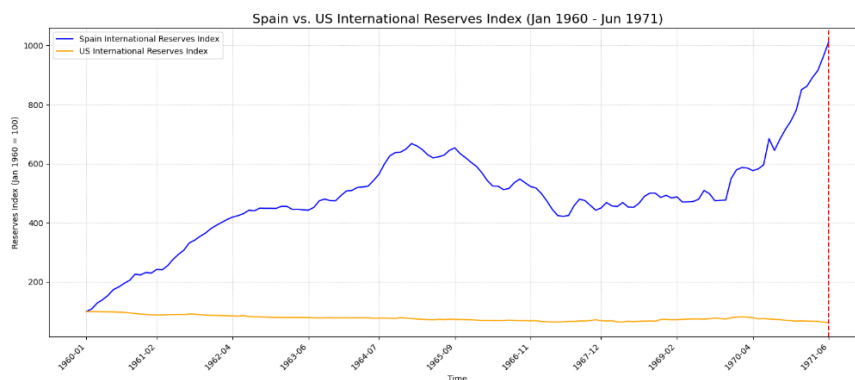
The 2x2 grid includes:

1. Nominal Exchange Rate Growth: Monthly percentage changes in Pesetas/USD. (left, up)
2. Nominal Exchange Rate Level: The raw exchange rate value. (right, up)
3. Real Exchange Rate Growth: Monthly percentage changes in the Real Exchange Rate (adjusted for inflation). (left, down)
4. Real Exchange Rate index (normalized to Dec 1990). (right, down)

Historical Context: A vertical dashed line is added at March 1973, representing the point when Spain moved away from the Bretton Woods fixed exchange rate system and allowed the peseta to float.

Note that higher figures are associated with Depreciation of the Peseta.

4.1.2, Spain and the US's International Reserves, 1960-1971, before US exit the B.W.



4.1.3, Answers to the Questions

Q1: Was the US dollar over- or undervalued in the Bretton Woods system?

The US dollar was overvalued under the Bretton Woods system. The real exchange rate index shows that the peseta was significantly undervalued relative to the dollar during the Bretton Woods period

(pre-1973), as the index stood well above 1 (around 2.5 in 1960, declining but still above 1.5 by 1973). This means it took more pesetas to buy a dollar than was justified by purchasing power, implying the dollar was overvalued relative to its true worth.

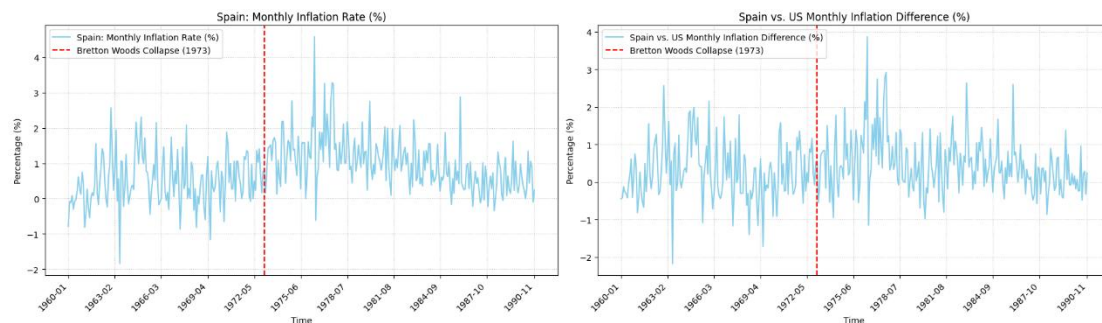
Q2: Using your graphs, explain one reason for the US's departure from Bretton Woods.

The international reserves graph shows that US reserves steadily declined from 1960 to 1971, falling to around 60% of their January 1960 level, while Spain's reserves grew dramatically (over 1000% by 1971). This reflects the so-called "Triffin Dilemma" — as the world's reserve currency, the US had to run persistent balance of payments deficits to supply dollars to the global economy, which continuously drained its gold reserves. By 1971, the US could no longer credibly maintain the dollar's convertibility to gold at \$35 per ounce, forcing Nixon to abandon the peg.

Section 4.2: Inflation and Industrial Production Graphs

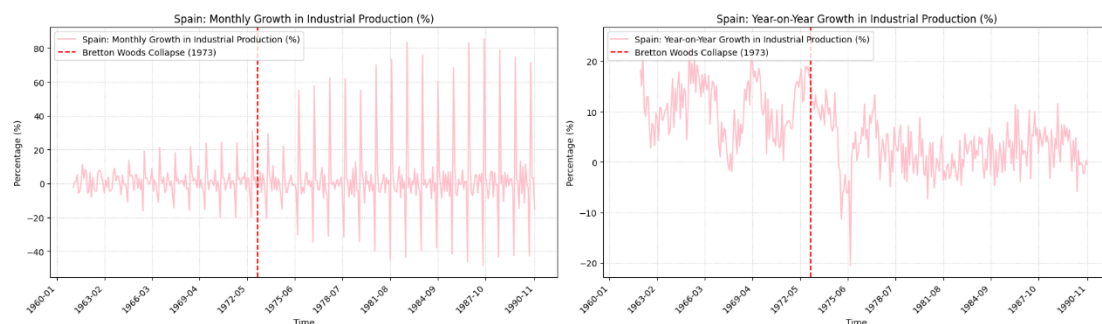
4.2.1, Monthly Inflation of Spain and Difference between US's and Spain's Monthly Inflation Rates

Spain vs. US Inflation Rates (1960-1990)



4.2.2, Monthly and Year-on-Year Growth in Spain's industrial production

Spain: Industrial Production Growth (1960-1990)



4.2.3, Answers to the Questions

Q1: Why are results for monthly versus 12-monthly growth so different? Which is more useful?

The monthly growth graph (left) shows extreme spikes reaching up to +80% and -40%, which are clearly driven by seasonal patterns in Spanish industrial production — factories regularly shut down and restart, creating artificial month-to-month swings that have nothing to do with the underlying economy. The 12-monthly graph (right) is far smoother, ranging between roughly -20% and +25%, because comparing each month to the same month a year earlier cancels out these seasonal effects. The 12-monthly measure is more useful as it reveals genuine economic trends, such as the clear slowdown in

industrial growth after 1973 and the deep dip around 1974-75 following the oil crisis.

Q2: Why use monthly data rather than quarterly or annual?

Monthly data gives us far more observations over the 1960-1990 period, which is essential for precisely identifying when volatility changed around the end of Bretton Woods in 1973. With quarterly or annual data, we would lose the ability to pinpoint exactly when structural breaks occurred. Monthly data also better captures short-run fluctuations in inflation (visible in the left inflation graph) and exchange rates that would be averaged away in lower-frequency data.

Section 4.3: Comparison Statistics & Visualization

4.3.1, Standard Deviations for Pre and Post Bretton Woods Periods

This section calculates the standard deviation for key economic indicators during two distinct periods:

- **Pre-Bretton Woods:** December 1959 to June 1971
- **Post-Bretton Woods:** March 1973 to December 1990

The indicators analyzed are:

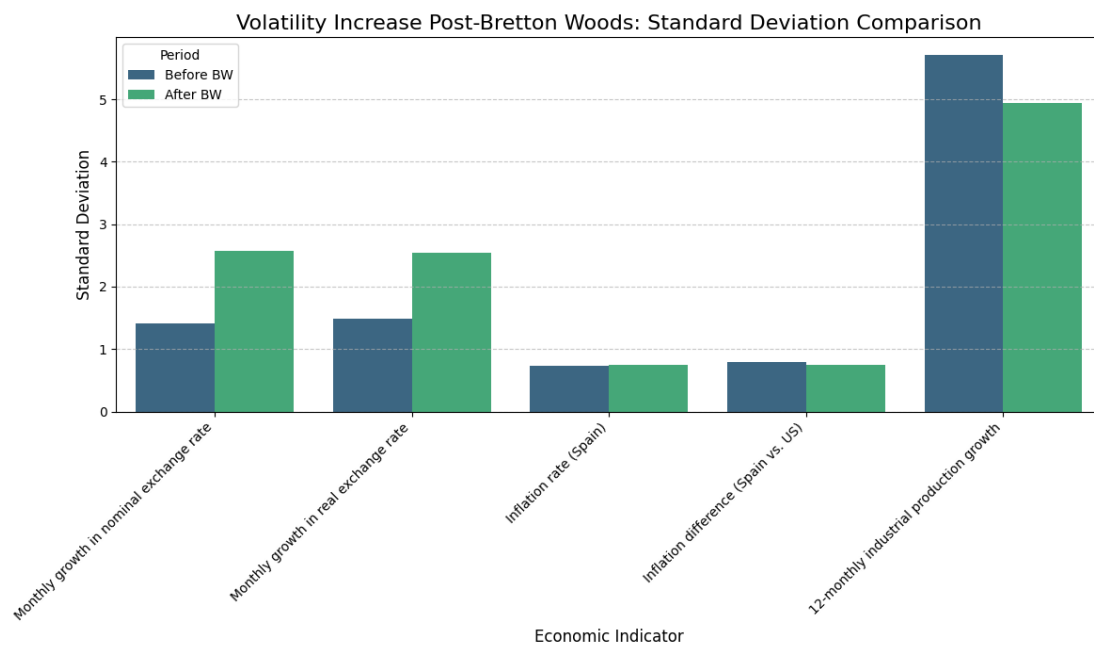
- Monthly % growth of the nominal exchange rate
- Monthly % growth of the real exchange rate
- Monthly inflation rate for Spain
- Difference between Spain and US monthly inflation
- 12-monthly industrial production growth

These figures are output in a contingency table, and later on visualized in a bar chart in 4.3.2.

Standard Deviations Before and After Bretton Woods for Spain:				
	Variable	Std Dev Before BW (until 06/1971)	Std Dev After BW (from 03/1973)	Ratio (Before/After)
0	Monthly growth in nominal exchange rate	1.4188	2.5733	0.5513
1	Monthly growth in real exchange rate	1.4807	2.5364	0.5838
2	Inflation rate (Spain)	0.7299	0.7419	0.9838
3	Inflation difference (Spain vs. US)	0.7883	0.7482	1.0536
4	12-monthly industrial production growth	5.7063	4.9363	1.1560

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4.3.2. Visualizing the Standard Deviations



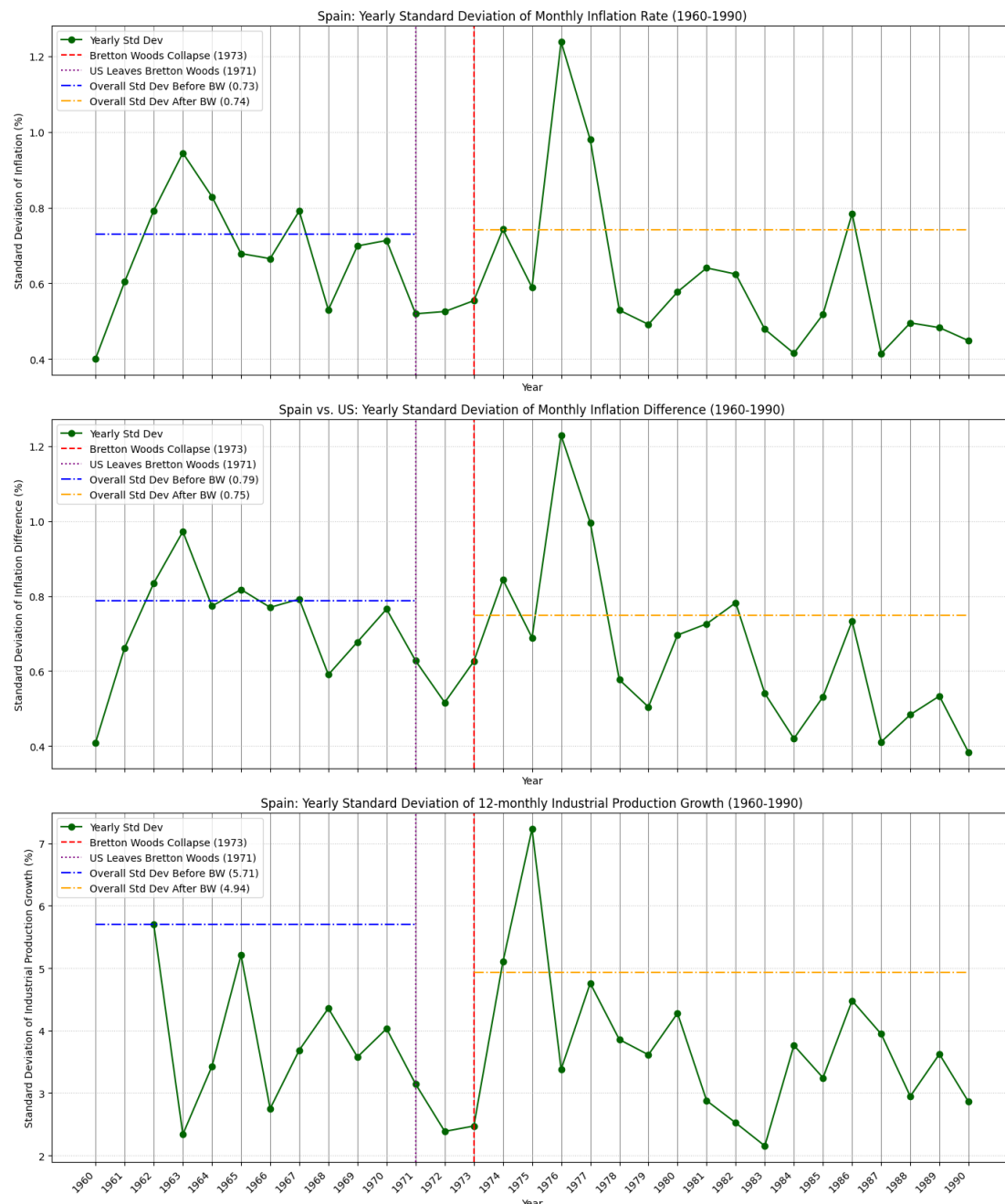
Q: Why exclude data from 07/1971 - 02/1973?

This period is a transition phase between the two regimes — the US suspended gold convertibility in August 1971 but countries had not yet fully moved to floating exchange rates. The Smithsonian Agreement of December 1971 attempted to maintain a modified fixed rate system, which only fully collapsed in March 1973. Including this transitional period would blur the comparison between the two distinct regimes, as exchange rates were neither fully fixed nor freely floating. Excluding it gives a cleaner before-and-after comparison.

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4.3.3, Volatility of Spain's the inflation rate, difference in the two countries' inflation rates, and the 12-monthly industrial production growth of Spain

Yearly Standard Deviations Before and After Bretton Woods (1960-1990)



Each plot includes:

1. **Yearly Standard Deviation:** One point per year representing the dispersion of the plotted variable within that specific year.
2. **Period Benchmarks:** Horizontal lines representing the average standard deviation for the Pre-BW and Post-BW periods calculated previously.
3. **Bretton Woods Exit:** A vertical dashed red line standing at 1973 for Spain leaving BW; a vertical dashed purple line standing at 1971 for the US leaving BW.

Section 4.4: Regression Analysis

4.4.1, Standard Deviations of Variables Calculated across Half-Year Windows

This section calculates the standard deviation for key economic variables within specific 6-month windows (the 1st and 2nd halves of a year) from 1960 to 1990.

year	half_year	std_err	std_inflation_spread	std_inflation_diff	std_yr_mn_yr_cr_indprod	29	1976	2	0.706555	0.529708	0.642111	3.725974	
0	1962	1	0.558347	0.587396	0.564962	4.710191	30	1977	1	0.995389	0.878453	0.952391	3.281219
1	1962	2	1.073079	1.012134	1.095935	3.685054	31	1977	2	5.892305	1.139063	1.128199	3.216441
2	1963	1	1.286745	1.245621	1.279556	2.441701	32	1978	1	1.210410	0.448308	0.532354	4.501698
3	1963	2	0.614453	0.576007	0.620854	2.464016	33	1978	2	2.607050	0.645055	0.665055	3.287959
4	1964	1	0.389706	0.513965	0.394450	3.728721	34	1979	1	1.184827	0.365787	0.448957	4.325636
5	1964	2	0.502918	0.524290	0.516563	3.362523	35	1979	2	0.701167	0.628383	0.580720	3.159754
6	1965	1	0.913650	0.710052	0.920618	6.950616	36	1980	1	2.844441	0.820807	0.810166	4.346984
7	1965	2	0.752973	0.708187	0.770139	3.105688	37	1980	2	1.195644	0.235018	0.529528	4.374366
8	1966	1	0.748515	0.664950	0.756268	1.653063	38	1981	1	2.966401	0.825716	0.904710	2.796064
9	1966	2	0.837649	0.726392	0.848764	3.185039	39	1981	2	2.324603	0.434271	0.477900	3.216983
10	1967	1	0.779042	0.754987	0.780086	3.097590	40	1982	1	3.233317	0.468600	0.688484	1.948001
11	1967	2	6.050344	0.853375	0.853554	3.022452	41	1982	2	1.172556	0.760256	0.929981	3.187690
12	1968	1	0.549059	0.514306	0.551995	3.265241	42	1983	1	1.431658	0.523427	0.554083	1.395477
13	1968	2	0.658794	0.565029	0.655678	3.980741	43	1983	2	1.561167	0.411188	0.549836	2.648072
14	1969	1	0.817284	0.840240	0.812355	3.380938	44	1984	1	2.323657	0.424705	0.399303	3.091539
15	1969	2	0.304596	0.353537	0.307016	3.629461	45	1984	2	2.311814	0.444954	0.470952	4.554740
16	1970	1	0.517912	0.569205	0.516131	4.666785	46	1985	1	3.515218	0.688276	0.723940	2.682156
17	1970	2	0.800532	0.696004	0.812901	3.307945	47	1985	2	2.294936	0.322695	0.307777	3.093507
18	1971	1	0.608432	0.461907	0.613617	3.936375	48	1986	1	2.395902	1.022388	0.917507	5.621218
19	1971	2	0.940887	0.602969	0.660093	1.425052	49	1986	2	2.355669	0.476181	0.444363	3.077716
20	1972	1	1.138163	0.393562	0.440779	1.453676	50	1987	1	1.752525	0.338189	0.272125	4.876928
21	1972	2	0.881329	0.470377	0.460445	3.221332	51	1987	2	1.670624	0.504778	0.496777	2.722274
22	1973	1	1.667723	0.541806	0.566459	2.348601	52	1988	1	1.933522	0.402823	0.456183	1.518605
23	1973	2	1.036247	0.601752	0.731348	1.506437	53	1988	2	3.116570	0.533208	0.448521	3.578759
24	1974	1	2.193488	0.851934	0.989406	3.048693	54	1989	1	2.156382	0.404956	0.442976	4.228125
25	1974	2	1.055240	0.610803	0.678485	5.826098	55	1989	2	3.158341	0.575592	0.577278	3.119967
26	1975	1	0.685992	0.535579	0.660222	2.895172	56	1990	1	1.574261	0.342868	0.168778	3.925141
27	1975	2	0.892154	0.683865	0.777068	9.738845	57	1990	2	2.305379	0.542270	0.512078	0.987527

4.4.2, OLS Regressions

Here is constructed the relationship between real exchange rate volatility and other economic volatilities using Ordinary Least Squares (OLS) regressions. The 6-month standard deviations are used as our data points.

Independent Variable (X): 6-month standard deviation of Real Exchange Rate Growth.

Dependent Variables (Y):

1. 6-month standard deviation of Spain's Inflation
2. 6-month standard deviation of the Inflation Difference between the US and Spain
3. 6-month standard deviation of Year-on-Year Industrial Production Growth

1. 6-month standard deviation of Spain's Inflation

Regression 1: Std Dev of Spain's Inflation Rate vs. Std Dev of Real Exchange Rate Growth						
OLS Regression Results						
Dep. Variable:	std_inflation_spain	R-squared:	0.111			
Model:	OLS	Adj. R-squared:	0.095			
Method:	Least Squares	F-statistic:	7.002			
Date:	Sun, 29 Mar 2026	Prob (F-statistic):	0.0106			
Time:	08:55:38	Log-Likelihood:	1.7907			
No. Observations:	58	AIC:	0.4186			
DF Residuals:	56	BIC:	4.540			
DF Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.5076	0.053	9.521	0.000	0.401	0.614
std_gr_rER	0.0690	0.026	2.646	0.011	0.017	0.121
Omnibus:	25.998	Durbin-Watson:	1.589			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	49.171			
Skew:	1.441	Prob(JB):	2.10e-11			
Kurtosis:	6.470	Cond. No.	4.06			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified						

2. 6-month standard deviation of the Inflation Difference between the US and Spain

Regression 2: Std Dev of Inflation Difference vs. Std Dev of Real Exchange Rate Growth

OLS Regression Results

Dep. Variable:	std_inflation_diff	R-squared:	0.069			
Model:	OLS	Adj. R-squared:	0.052			
Method:	Least Squares	F-statistic:	4.119			
Date:	Sun, 29 Mar 2026	Prob (F-statistic):	0.0472			
Time:	08:55:38	Log-Likelihood:	-1.2402			
No. Observations:	58	AIC:	6.480			
Df Residuals:	56	BIC:	10.60			
Df Model:	1					
Covariance Type:	nonrobust					

	coef	std err	t	P> t	[0.025	0.975]
const	0.5686	0.056	10.122	0.000	0.456	0.681
std_gr_rER	0.0558	0.027	2.029	0.047	0.001	0.111

Omnibus:	13.133	Durbin-Watson:	1.331			
Prob(Omnibus):	0.001	Jarque-Bera (JB):	15.234			
Skew:	0.930	Prob(JB):	0.000492			
Kurtosis:	4.688	Cond. No.	4.06			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

3. 6-month standard deviation of Year-on-Year Industrial Production Growth

Regression 3: Std Dev of Spain's 12-monthly Industrial Production Growth vs. Std Dev of Real Exchange Rate Growth						
OLS Regression Results						
Dep. Variable:		std_yr_on_yr_gr_ind_prod	R-squared:		0.023	
Model:		OLS	Adj. R-squared:		0.006	
Method:		Least Squares	F-statistic:		1.334	
Date:		Sun, 29 Mar 2026	Prob (F-statistic):		0.253	
Time:		08:55:38	Log-Likelihood:		-101.09	
No. Observations:		58	AIC:		206.2	
Df Residuals:		56	BIC:		210.3	
Df Model:		1				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
const	3.7044	0.314	11.791	0.000	3.075	4.334
std_gr_rER	-0.1776	0.154	-1.155	0.253	-0.485	0.130
Omnibus:	33.489		Durbin-Watson:		2.021	
Prob(Omnibus):	0.000		Jarque-Bera (JB):		99.524	
Skew:	1.612		Prob(JB):		2.45e-22	
Kurtosis:	8.549		Cond. No.		4.06	
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

Comments on the Results:

1. An increase in the 6-month standard deviation of Real Exchange Rate Growth is associated with an increase in the 6-month standard deviation of Spain's inflation rate by 0.0690 on average. Both the constant and the slope estimates are significant statistically provided a 0.95 level of significance given the p-value.
2. An increase in the 6-month standard deviation of Real Exchange Rate Growth is associated with an increase in the 6-month standard deviation of the Inflation Difference between the US and Spain by 0.0558 on average. Both the constant and

the slope estimates are significant statistically provided a 0.95 level of significance given the p-value.

3. An increase in the 6-month standard deviation of Real Exchange Rate Growth is associated with a decline in the 6-month standard deviation of Year-on-Year Industrial Production Growth by 0.1776 on average. Both the constant and the slope estimates are significant statistically provided a 0.95 level of significance given the p-value.

Looking at the results, it may be concluded that though the linear OLS suggests with statistical significance that greater volatility in real exchange rate growth is associated with more volatility in Spain's inflation and the difference between the US's and Spain's inflation rates, as well as less volatility in Spain's industrial production, the estimated slope coefficients are rather minimal. This might leave it questionable whether a stable real exchange rate growth does matter to other economic performances, despite the predictions with statistical significance.

R-squared values are no larger than 0.15, formally meaning each bivariate linear regression with the volatility of real exchange rate growth as the independent variable (X) provide little explanation in the change in volatility in the dependent variables (Y's). However, considering the vast number of omitted variables and real-life complexity, these R-squared values might as well be regarded as natural, not necessarily judging the fitness of the models. What is concerning is that the standard errors of the two estimated coefficients in the 2nd and the 3rd regressions seem rather large, reaching approx. 50% and 100% of the estimated coefficients by approximation, adding uncertainty to the constructed correlations.

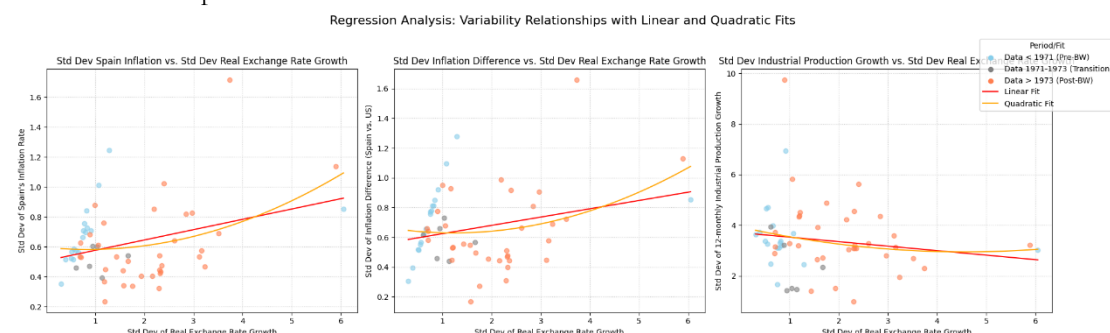
4.4.3, Visualizing the Regressions

This section visualizes the relationship between Real Exchange Rate volatility and other economic volatilities using 6-month standard deviations.

Colors:

1. Blue: Pre-Bretton Woods (Up to June 1971)
2. Coral Orange: Post-Bretton Woods (From March 1973)
3. Grey: Transition Period (July 1971 - February 1973)

Each plot includes both a linear and a quadratic (second-degree) line of best fit to explore the nature of the relationship.



Answer to the Question:

Q: What regressions have been implicitly run to produce the linear and quadratic lines of best fit?

The linear line of best fit implicitly runs an OLS regression of the form: $Y = \beta_0 + \beta_1 X$, where Y is the dependent variable (e.g. standard deviation of inflation) and X is the standard deviation of real exchange

rate growth. The quadratic line of best fit implicitly runs: $Y = \beta_0 + \beta_1 X + \beta_2 X^2$, where X^2 is the squared term of the independent variable. This is still a linear regression in the coefficients — it is estimated by OLS by simply adding X^2 as an additional regressor alongside X .

Section 5: Extensions

Section 5.1: Studying Levels vs. Volatility

5.1.1. Average Level Analysis: Pre and Post Bretton Woods for Spain

This section calculates the mean (average) levels of the following key economic indicators for the two primary regimes:

- **Pre-Bretton Woods:** Dec 1959 - Jun 1971
- **Post-Bretton Woods:** Mar 1973 - Dec 1990

Variables Analyzed:

- Nominal exchange rate level (Pesetas/USD)
- Real exchange rate index
- Monthly inflation rate for Spain
- 12-monthly industrial production growth

The results are presented in a formatted table with pre/post columns and ratio/difference columns, followed by bar charts for visual comparison.

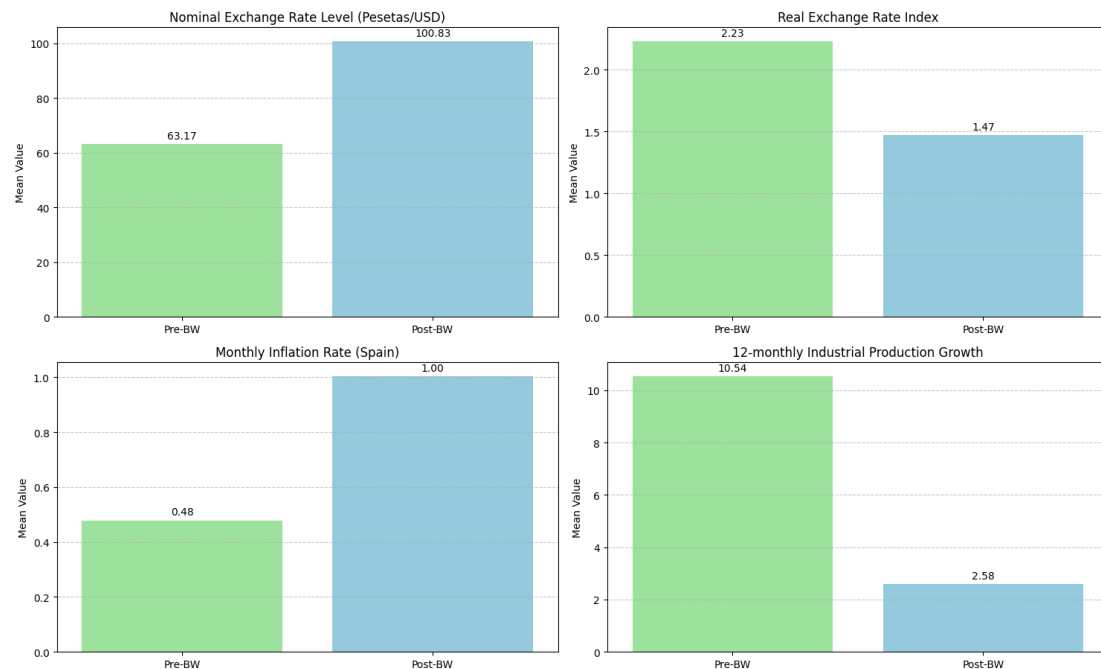
Average Levels of Key Economic Indicators (Pre vs. Post Bretton Woods):					
	Variable	Mean Pre-BW (Dec 1959 – Jun 1971)	Mean Post-BW (Mar 1973 – Dec 1990)	Difference (Post – Pre)	Ratio (Post / Pre)
0	Nominal Exchange Rate Level (Pesetas/USD)	63.1655	100.8266	37.6612	1.5962
1	Real Exchange Rate Index	2.2309	1.4728	-0.7581	0.6602
2	Monthly Inflation Rate (Spain)	0.4789	1.0037	0.5247	2.0957
3	12-monthly Industrial Production Growth	10.5399	2.5836	-7.9563	0.2451

(Visualizations on the next page)

5.2.2. Visualizing Average Levels (Pre vs. Post Bretton Woods)

This plot visualizes the average levels calculated in the previous step, providing a quick comparison of economic conditions before and after the Bretton Woods system.

Average Levels of Key Economic Indicators (Pre vs. Post Bretton Woods)



5.1.3. Sensitivity Analysis: Progressively Longer Windows around 1973

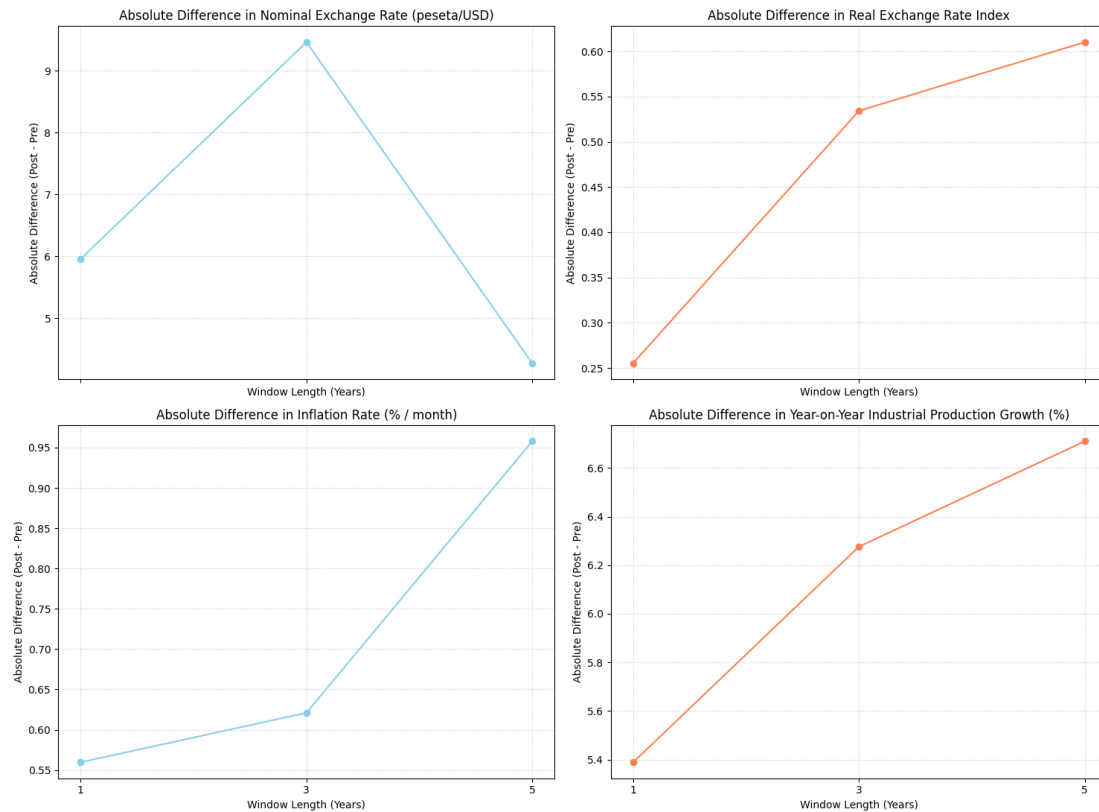
This section examines the robustness of our average level comparisons by using progressively longer time windows around the March 1973 Bretton Woods exit date. This helps to see how the choice of period length influences the observed differences between the pre and post Bretton Woods eras.

We calculate average levels for windows of plus/minus 1, 3, or 5 years around the transition (1973). A table summarizes these results, and a line plot shows the trend of post-1973 averages as the window size increases. Meaning: the mean values for different lengths of time windows are shown in a contingency table. Then, we demonstrate the change in "the difference between the mean values" with "length of time window" in a matrix of 4 line-plots.

Contingency Table of Average Levels Before and After Spain's Bretton Woods Exit (March 1973) :						
Window (Years)	1		3		5	
Period	Post-Bretton Woods (Avg)	Pre-Bretton Woods (Avg)	Post-Bretton Woods (Avg)	Pre-Bretton Woods (Avg)	Post-Bretton Woods (Avg)	Pre-Bretton Woods (Avg)
Variable						
Inflation Rate (% / month)	1.2288	0.6691	1.2818	0.6609	1.4556	0.4977
Nominal Exchange Rate (peseta/USD)	57.7088	63.6660	58.0237	67.4842	64.2184	68.4905
Real Exchange Rate Index	1.5809	1.8363	1.5050	2.0391	1.4861	2.0962
Year-on-Year Industrial Production Growth (%)	10.6621	16.0512	3.9222	10.1985	4.3680	11.0780

Visualization: the change in **absolute differences between pre and post averages** of the four variables with window length (1, 3, or 5 years):

Absolute Difference in Average Levels (Post BW - Pre BW) by Window Length



5.1.4. Answers to the Questions

Q1: Is there evidence that average levels changed markedly around the end of Bretton Woods? Nominal and Real Exchange Rates?

- Yes, the nominal exchange rate changed notably.

The average level of pesetas per USD rose from 63.17 pre-Bretton Woods to 100.83 post-Bretton Woods — a 60% increase (ratio of 1.60), reflecting the peseta's depreciation once it was free to float. The real exchange rate index fell from an average of 2.23 to 1.47 (ratio of 0.66), meaning the peseta appreciated in real terms post-1973, likely driven by Spain's higher inflation eating into the nominal depreciation.

- Inflation:

Average monthly inflation more than doubled from 0.48% pre-Bretton Woods to 1.00% post-Bretton Woods (ratio of 2.10), suggesting inflation became significantly more serious after the fixed exchange rate was removed.

- Industrial Production Growth:

The most dramatic change — average 12-monthly industrial production growth collapsed from 10.54% pre-Bretton Woods to just 2.58% post-Bretton Woods (ratio of 0.25).

Q2: How do results change with longer periods?

Looking at the windowed comparison table (± 1 , ± 3 , ± 5 years around 1973):

The nominal exchange rate pre-average rises from 63.67 (± 1 year) to 68.49 (± 5 years), while the post-average rises from 57.64 to 63.86, meaning the gap between pre and post actually narrows with longer

windows (despite the abrupt rise for the 3-year window length: perhaps exogenous events impacted Spain significantly).

Inflation's post-average increases from 1.23 (± 1 year) to 1.46 (± 5 years), suggesting inflation kept rising for several years after 1973, so longer windows capture more of this increase.

Industrial production growth post-average falls from 10.66 (± 1 year) to 4.37 (± 5 years), meaning the growth slow-down looks worse over longer windows as Spain's post-oil-crisis recession is captured.

Overall, results become more sensitive to events far from 1973 as windows widen, making the before/after comparison less 'clean'.

Q3: What issues are there with using longer periods to infer causality?

Confounding events — the 1973 and 1979 oil shocks, Spain's political transition to democracy after Franco's death in 1975, and global recessions, all occurred in the post-1973 period. Longer windows capture these events, making it impossible to isolate the effect of the exchange rate regime change alone. Meanwhile, the Spanish Miracle — Spain's unusually high industrial production growth pre-1973 (averaging 10.5%) -- reflected a very unique phase of growth, not just the impacts from the fixed exchange rate. Attributing the post-1973 slowdown to exchange rate flexibility is therefore misleading. Confounders over long periods, such as structural changes in the Spanish economy (industrialisation, EU accession process, labour market changes), thus similarly affect all variables independently of exchange rate.

Last but not least, these could still be risks of Reverse Causality: Spain's higher post-1973 inflation may have caused the peseta to depreciate, rather than exchange rate volatility causing inflation.

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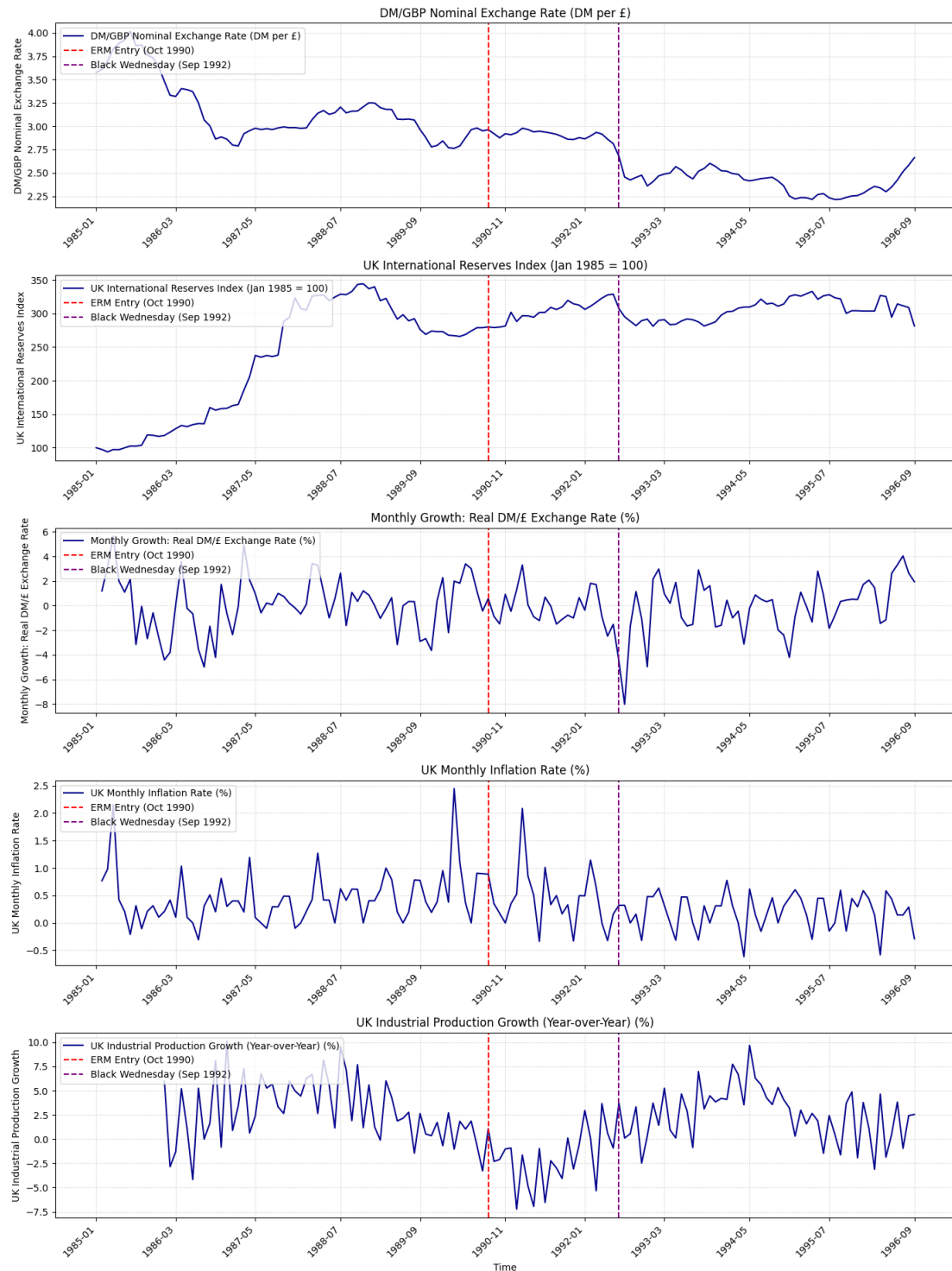
Section 5.2: Black Wednesday: The UK Joining & Leaving the ERM

5.1.1, [Loading Data/Creating Variables](#), see code.

5.2.2, Key UK Economic Indicators (1985-1997)

Data was collected from 1985 (5 years before the UK joined the ERM) to 1997 (5 years after the UK left the ERM). Note here, higher ER indices mean appreciation of the GBP.

Key UK Economic Indicators (1985-1997)



5.2.3, Standard Deviation Analysis: Pre-ERM, ERM, and Post-ERM Periods

Standard Deviations Before, During, and After ERM for the UK:						
	Variable	Std Dev Pre-ERM (Jan 1985 – Sep 1990)	Std Dev ERM (Oct 1990 – Aug 1992)	Std Dev Post-ERM (Sep 1992 – Dec 1997)	Ratio (ERM/Pre-ERM)	Ratio (Post-ERM/ERM)
0	DM/GBP Nominal Exchange Rate Growth	2.1291	1.0216	2.2981	0.4798	2.2494
1	DM/GBP Real Exchange Rate Growth	2.2593	1.3521	2.2841	0.5985	1.6892
2	UK Monthly Inflation Rate	0.4786	0.5392	0.3241	1.1266	0.6011
3	UK Industrial Production Growth (YoY)	3.2838	2.9143	2.7376	0.8875	0.9394
4	UK/USD Nominal Exchange Rate Growth	2.9866	2.9308	2.7411	0.9813	0.9353

5.2.4, Average Levels Analysis: Pre-ERM, ERM, and Post-ERM Periods

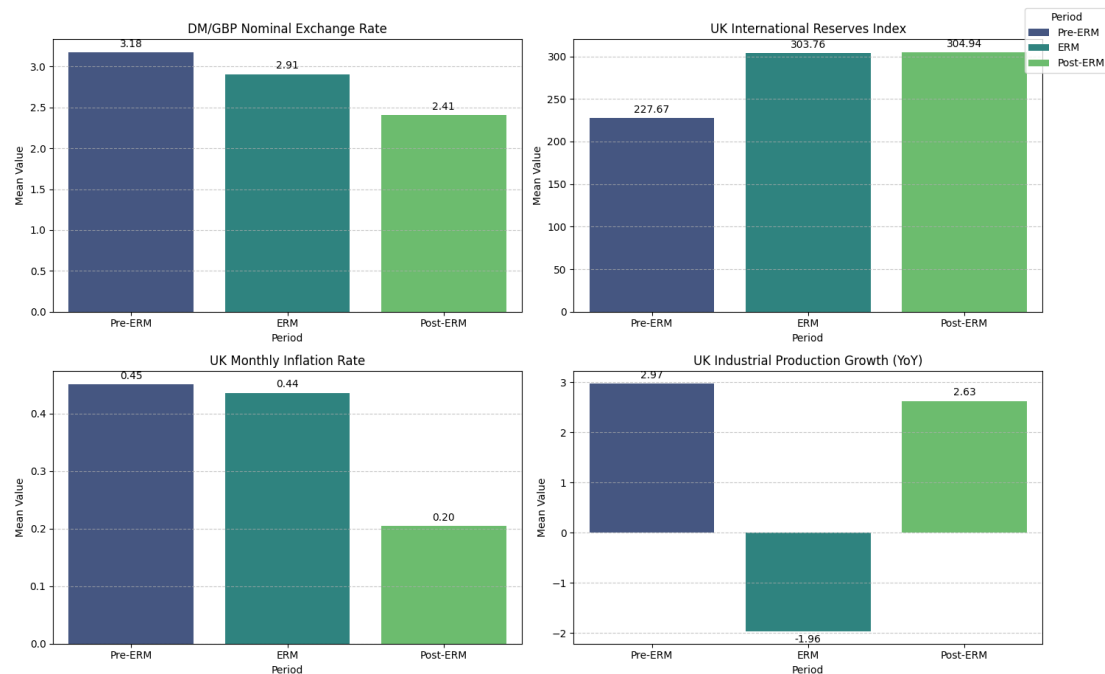
Average Levels Before, During, and After ERM for the UK:								
	Variable	Mean Pre-ERM (Jan 1985 – Sep 1990)	Mean ERM (Oct 1990 – Aug 1992)	Mean Post-ERM (Sep 1992 – Dec 1997)	Ratio (ERM/Pre-ERM)	Ratio (Post-ERM/ERM)	Diff (ERM – Pre-ERM)	Diff (Post-ERM – ERM)
0	DM/GBP Nominal Exchange Rate	3.1752	2.9082	2.4069	0.9159	0.8276	-0.2670	-0.5013
1	UK International Reserves Index	227.6718	303.7637	304.9446	1.3342	1.0039	76.0919	1.1809
2	UK Monthly Inflation Rate	0.4514	0.4364	0.2050	0.9668	0.4697	-0.0150	-0.2314
3	UK Industrial Production Growth (YoY)	2.9748	-1.9637	2.6296	-0.6601	-1.3391	-4.9385	4.5933

(for visualizations of these average levels, see the next page)

Visualization of Average Levels (Pre-ERM, ERM, and Post-ERM Periods):

This bar chart visualizes the average levels calculated in the previous step, providing a clear comparison of economic conditions before, during, and after the UK's membership in the ERM.

Average Levels of Key UK Economic Indicators (Pre-ERM, ERM, Post-ERM)



5.2.5. Answers to the Questions

Q1: What was a key reason for why the UK chose to join the ERM?

It is perceived that Inflation was a key driver for the UK to join the ERM: UK's inflation was hitting high during 1990 with a maximal rate of 2.5% PER MONTH. Given that the Deutsche Mark was a stable standard currency held by Germany's Bundesbank to counter inflationary risks in the newly reunified Germany, pegging the GBP to the Deutsche Mark could stabilise the UK's domestic inflation through a series of operations. While inflation stayed high and the GBP was held to not depreciate naturally, the market would see that the GBP was overvalued and attempt to sell GBP in exchange for currencies that did better in value storing, causing depreciatory pressure on GBP. To maintain a stable exchange rate, the Bank of England must buy in GBP to exert appreciatory pressure. This would reduce money supply in Britain and counter domestic inflation. However, after joining the ERM, the average level of inflation rate seemed to decrease only marginally (from 0.45% monthly average before to 0.44% monthly average during the ERM era).

Q2: What factors led to Soros and speculators to run on the pound?

We suggest two reasons:

1. Trend in Sterling's exchange rate and valuation provided the chance.

Across the 1985-1996 period, it was observable that the pound was roughly following a dropping trend, and during the last few months in the ERM this trend was exceptionally potent, reflecting the market's recognition that the pound was overvalued under the managed exchange rate system. Besides, even after joining the

ERM, inflation in the UK still recorded some extraordinary values, exceeding 2.0% per month around early 1991 and exceeding 1.0% per month during the last few months in the ERM. These could have enhanced the market's suspicion in the pound's value-storing ability and confirmed the recognition that the pound was overvalued. Utilizing such mindsets, Soros could excessively sell out the pound and smash on the pound's exchange rate to increase the suspicion on Sterling's value, causing the market to doubt the pound's value and sell out the pound collectively.

2. Keeping a managed exchange rate was essentially Unnatural for the scenario the UK was faced with.

Industrial production in the UK seemed terrible during the ERM years (average year-on-year growth was negative 1.96), and it would be natural to allow the pound to drop while lowering interest rates to stimulate domestic production by enhancing British exports' price competitiveness and encouraging borrowing. However, with a managed exchange rate, the Bank of England had to buy in the pound to counter depreciatory pressure, reducing money supply; meanwhile, it could not freely lower interest rates because of concerns of hot money outflows (to countries where rates were higher for saving, e.g., the newly reunited Deutschland), which would include selling pounds to cause more depreciatory pressure. It thus seemed not a sustainable choice to persistently maintain the pegged exchange rate in sacrifice of limited foreign reserves and domestic economy.

Q3: Is there evidence of changes in volatility after joining/leaving the ERM?

Looking at the Standard Deviation Table (5.2.3):

1. **For DM/GBP exchange rate growth itself, there was evidence of changes in volatility after joining and leaving the ERM.**

It was observable that joining the managed exchange rate system reduced the dispersion in the growth rates of nominal and real DM/GBP exchange rates: the ERM/pre-ERM ratios for the standard deviations of the two growth rates were respectively 0.4798 and 0.5985. Leaving the ERM made the growth rates of nominal and real DM/GBP figures bounce volatile again: the post-ERM/ERM ratios for the two growth rates were 2.249 and 1.689.

2. **Joining the ERM didn't seem to cause significant impacts on the volatility of key economic performances such as inflation and industrial production.** The standard deviations of inflation rates and year-on-year industrial production growth compared similar to their pre-ERM levels during the ERM period (ERM/pre-ERM ratios for the standard deviations were 1.1266 and 0.8875, both close to 1). While leaving the ERM did not seem to impact industrial production much (post-ERM/ERM standard deviation ratio was 0.9394, still close to 1), it seemed conversely that leaving the ERM stabilized inflation: the post-ERM/ERM standard deviation ratio for UK's monthly inflation was 0.6011, almost halving the volatility of inflation.
3. Nor did external exchange rates, like the UK/USD rate, seem impacted because of joining or leaving the ERM. Both the ERM/pre-ERM and the post-ERM/ERM standard deviation ratios of the growth of UK/USD nominal exchange rate were both very close to 1.

Q4: Changes in levels following Black Wednesday?

From the line graphs:

Nominal DM/GBP exchange rate was allowed to fall sharply, so did the real DM/GBP exchange rate, confirming an essential decline in the pound's value or purchasing power.

Combining the Average Level Bar Plots and the Standard Deviation Table:

Inflation was stabilized on a much lower average level (0.20% per month) compared to the ERM era (0.44% per month), also considering a much lower level of volatility (as in the answer to Q3).

Industrial Production rebounded. Post-ERM average levelled at 2.63 index value as a definite reverse to the -1.96 index value during the ERM era, while it has been analyzed that volatility did not significantly change.

Q5: What concerns do you have about drawing conclusions from this period?

1. **The ERM era was too short:** only 2 years of data (Oct 1990 to Sep 1992) makes it very difficult to draw solid conclusions. Standard deviations and averages based on just 24 months are highly sensitive to individual outliers: the period could be coincidently a particularly weak time for the British economy, and the economy was facing the dilemma of keeping exchange rate stable against restoring domestic economic performances. *All these specific cases could have made this historical era 'special' and ungeneralisable for analyses regarding the importance of exchange rates though we seem to conclude that exchange rates are not in fact so important as they used to be reckoned. It might be that in other cases (like Spain's cases), there would be some remarkable changes that are potentially associated with exchange rate volatility.*
2. **Confounding factors:** the ERM period coincided with a global recession and the Gulf War (1990-91), which independently depressed UK industrial production. It is hard to separate the effect of the exchange rate peg from these external shocks. German reunification also caused abnormally high German interest rates, which forced the conflict between UK's domestic economic restoration and exchange rate requirements. This was a very unique event that seems unlikely to repeat.
3. **Policy would have changed after the Black Wednesday:** the UK adopted official inflation targeting to reduced inflation volatility on its own. This makes it hard to attribute the post-ERM inflation improvement solely to exchange rate flexibility.